

Project 3 – Hosting a Website on Raspberry Pi

This project demonstrates how to host a simple web server on a Raspberry Pi using Python's built-in HTTP server module, and serve a custom HTML page over the local network.

Steps

1. Find the Raspberry Pi's IP address

```
ifconfig
```

This displays network interface details (e.g., `eth0`) including the device's MAC address and network stats, which help identify the Pi on the local network.

2. Switch to root user

```
sudo -s
```

Elevates the current user to root, needed to bind the server to port 80 (ports below 1024 require root privileges).

3. Start the web server

```
python -m http.server 80
```

- **Python** – runs a lightweight built-in HTTP server (no extra installation needed)
- **HTTP** – the protocol used for client-server communication
- **Port 80/8080** – the port the server listens on; 80 is the default web port, so no `:port` is needed in the browser URL

Once running, the terminal logs every incoming request (GET requests, status codes like 200/404, timestamps, and client IP).

4. Create and edit the webpage

```
touch index.html      # create a new  
empty file
```

```
ls                # list
files/folders to confirm it was
created
nano index.html   # open the
file in the nano text editor
```

Inside `nano`, add your HTML content, then save and exit with:

```
Ctrl + X → Y → Enter
```

Result

With the server running and `index.html` in the same directory, visiting the Pi's IP address (e.g., `http://<pi-ip-address>/`) from any device on the same network loads the hosted page.

Requirements

- Raspberry Pi with Raspberry Pi OS
- Python 3 (pre-installed on most Raspberry Pi OS versions)
- Terminal access (local or via SSH)

How to Run

1. Open a terminal on the Raspberry Pi (or SSH into it).
2. Run `ifconfig` to note the Pi's IP address.
3. Run `sudo -s` to switch to root.
4. Navigate to the folder containing your `index.html` file.
5. Run `python -m http.server 80`.
6. On any device connected to the same network, open a browser and go to `http://<pi-ip-address>/`.